

TL044M, TL044C QUAD LOW-POWER OPERATIONAL AMPLIFIERS

D1662, SEPTEMBER 1973—REVISED JUNE 1988

- Very Low Power Consumption
- Typical Power Dissipation with ± 2 -V Supplies . . . 340 μ W
- Low Input Bias and Offset Currents
- Output Short-Circuit Protection
- Low Input Offset Voltage
- Internal Frequency Compensation
- Latch-Up-Free Operation
- Power Applied in Pairs

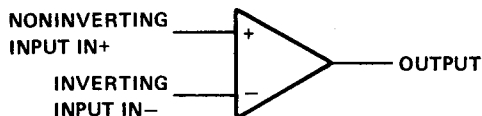
TL044M IS NOT RECOMMENDED FOR NEW DESIGNS.

description

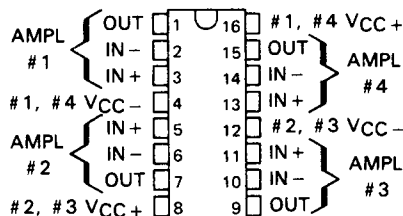
The TL044 is a quad low-power operational amplifier designed to replace higher-power devices in many applications without sacrificing system performance. High input impedance, low supply currents, and low equivalent input noise voltage over a wide range of operating supply voltages result in an extremely versatile operational amplifier for use in a variety of analog applications including battery-operated circuits. Internal frequency compensation, absence of latch-up, high slew rate, and output short-circuit protection assure ease of use. Power may be applied separately to Section A (amplifiers 1 and 4) or Section B (amplifiers 2 and 3) while the other pair remains unpowered.

The TL044M is characterized for operation over the full military temperature range of -55°C to 125°C ; the TL044C is characterized for operation from 0°C to 70°C .

symbol (each amplifier)

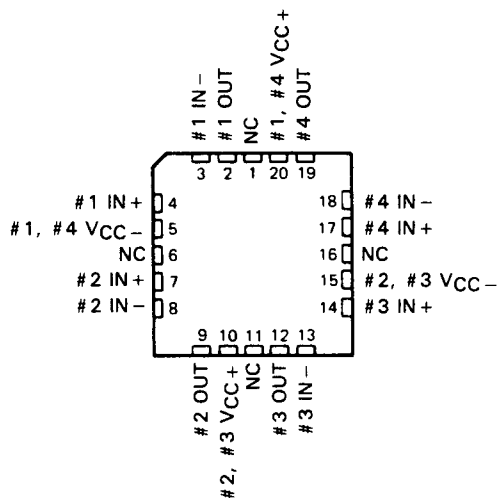


**TL044M . . . J OR W DUAL-IN-LINE PACKAGE
TL044C . . . J OR N PACKAGE
(TOP VIEW)**



Pins 4 and 12 are internally connected together in the N package only.

**TL044M . . . FK PACKAGE
(TOP VIEW)**



NC—No internal connection

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AVAILABLE OPTIONS

T _A	V _{IO} MAX AT 25°C	PACKAGE			
		CHIP CARRIER (FK)	CERAMIC DIP (J)	PLASTIC DIP (N)	FLAT PACK (W)
0°C to 70°C	5 mV	—	TL044CJ	TL044CN	—
–55°C to 125°C	5 mV	TL044MFK	TL044MJ	—	TL044MW

absolute maximum ratings over operating free-air temperature range (unless otherwise noted)

	TL044M	TL044C	UNIT
Supply voltage V _{CC} + (see Note 1)	22	18	V
Supply voltage V _{CC} – (see Note 1)	–22	–18	V
Differential input voltage (see Note 2)	±30	±30	V
Input voltage (any input, see Notes 1 and 3)	±15	±15	V
Duration of output short-circuit (see Note 4)	unlimited	unlimited	
Continuous total dissipation	See Dissipation Rating Table		
Operating free-air temperature range	–55 to 125	0 to 70	°C
Storage temperature range	–65 to 150	–65 to 150	°C
Case temperature for 60 seconds	FK package	260	°C
Lead temperature 1,6 mm (1/16 inch) from case for 60 seconds	J or W package	300	°C
Lead temperature 1,6 mm (1/16 inch) from case for 10 seconds	N package	260	°C

- NOTES: 1. All voltage values, unless otherwise noted, are with respect to the midpoint between V_{CC} + and V_{CC} –.
2. Differential voltages are at the noninverting input terminal with respect to the inverting input terminal.
3. The magnitude of the input voltage must never exceed the magnitude of the supply voltage or 15 V, whichever is less.
4. The output may be shorted to ground or either power supply. For the TL044M only, the unlimited duration of the short-circuit applies at (or below) 125°C case temperature or 85°C free-air temperature.

DISSIPATION RATING TABLE

PACKAGE	T _A ≤ 25°C POWER RATING	DERATING FACTOR	DERATE ABOVE T _A	T _A = 70°C POWER RATING	T _A = 125°C POWER RATING
FK	680 mW	11.0 mW/°C	88°C	680 mW	275 mW
J (TL044M)	680 mW	11.0 mW/°C	88°C	680 mW	275 mW
J (TL044C)	680 mW	8.2 mW/°C	67°C	656 mW	—
N	680 mW	N/A	N/A	680 mW	—
W	680 mW	8.0 mW/°C	65°C	640 mW	200 mW

TL044M, TL044C

QUAD LOW-POWER OPERATIONAL AMPLIFIERS

electrical characteristics at specified free-air temperature, $V_{CC+} = 15\text{ V}$, $V_{CC-} = -15\text{ V}$

PARAMETER		TEST CONDITIONS†	TL044M			TL044C			UNIT
			MIN	TYP	MAX	MIN	TYP	MAX	
V _{IO}	Input offset voltage	V _O = 0, R _S = 50 Ω	25 °C	1	5	1	5	mV	
			Full range	6			7.5		
I _{IO}	Input offset current	V _O = 0	25 °C	5	40	15	80	nA	
			Full range	100			200		
I _{IB}	Input bias current	V _O = 0	25 °C	50	100	100	250	nA	
			Full range	250			400		
V _{ICR}	Common-mode input voltage range		25 °C	± 12	± 13	± 12	± 13	V	
			Full range	± 12			± 12		
V _{OPP}	Maximum peak-to-peak output voltage swing	R _L = 10 kΩ	25 °C	20	26	20	26	V	
		R _L ≥ 10 kΩ	Full range	20			20		
A _{VD}	Large-signal differential voltage amplification	R _L ≥ 10 kΩ, V _O = ± 10 V	25 °C	72	86	60	80	dB	
			Full range	66			60		
B ₁	Unity-gain bandwidth		25 °C	0.5			0.5	MHz	
CMRR	Common-mode rejection ratio	V _{IC} = V _{ICR} min, V _O = 0, R _S = 50 Ω	25 °C	60	72	60	72	dB	
			Full range	60			60		
k _{SVS}	Supply voltage sensitivity (ΔV _{IO} /ΔV _{CC})	V _{CC} = ± 9 V to ± 15 V, V _O = 0, R _S = 50 Ω	25 °C	30	150	30	200	μV/V	
			Full range	150			200		
V _n	Equivalent input noise voltage	A _{VD} = 20 dB, B = 1 Hz, f = 1 kHz	25 °C	50			50	nV/√ Hz	
I _{OS}	Short-circuit output current		25 °C	± 6			± 6	mA	
I _{CC}	Supply current (four amplifiers)	No load, V _O = 0 V	25 °C	250	400	250	500	μA	
			Full range	400			500		
P _D	Total dissipation (four amplifiers)	No load, V _O = 0 V	25 °C	7.5	12	7.5	15	mW	
			Full range	12			15		

† All characteristics are measured under open-loop conditions with zero common-mode input voltage, unless otherwise specified. Full range for TL044M is -55°C to 125°C and for TL044C is 0°C to 70°C .

operating characteristics, $V_{CC+} = 15\text{ V}$, $V_{CC-} = -15\text{ V}$, $T_A = 25^\circ\text{C}$

PARAMETER		TEST CONDITIONS		TL044M			TL044C			UNIT
				MIN	TYP	MAX	MIN	TYP	MAX	
t _r	Rise time	V _I = 20 mV,	R _L = 10 kΩ,	0.3			0.3			μs
	Overshoot factor	C _L = 100 pF,	See Figure 1	5%			5%			
SR	Slew rate at unity gain	V _I = 10 V,	R _L = 10 kΩ,	0.5			0.5			V/μs
		C _L = 100 pF,	See Figure 1							